MODULE 1 · LECTURE 6

Database Management Systems: Introduction, Types and Applications in Agriculture

Why databases replace scattered files, how data is modelled and organised, relational tables, keys and SQL, and the national agricultural databases built on them.

Learning objectives

After this lecture you should be able to:

1. Define data, database and database management system (DBMS), and list the components and users of a database system.
2. Compare a DBMS with a traditional file system.
3. Explain the three-schema architecture and data independence.
4. Describe the hierarchical, network, relational, object-oriented and NoSQL data models.
5. Use relational terms, keys, ER diagrams and normalisation to design a simple farm database.
6. Write basic SQL queries and explain transactions, security and privacy.
7. Describe the use of databases in Indian agriculture.

Scope: Lecture 5 showed how to keep farm records in a spreadsheet. This lecture explains why large, shared records need a database. The Internet, through which most databases are reached, follows in Lecture 7; national digital agriculture platforms are covered in Lectures 30 and 31.

1. Data, databases and DBMS

Term	Meaning	Example
Data	Raw facts	Farmer name, plot area, soil pH reading
Database	An organised collection of related data, stored so that it can be shared and used for many purposes	Soil-test records of all farmers in a district
Database management system (DBMS)	Software that creates, stores, retrieves, updates and protects databases, and controls who can use them	MS Access, MySQL, PostgreSQL, Oracle
Database system	The database, the DBMS, the hardware, the procedures and the people together	A state soil-testing laboratory network

Users of a database system

- **Database administrator (DBA)** — designs storage, controls access, plans backups and tunes performance.
- **Database designers** — decide the structure: tables, fields, keys and relationships.
- **Application programmers** — write the forms, apps and reports that use the database.
- **End users** — farmers, extension officers, scientists and managers who enter and query data, usually through an app or website.

2. Why not just use files?

Before databases, each office kept its own files — registers, spreadsheets and folders. The problems that follow are the reasons DBMS were developed (Elmasri & Navathe, 2016; Silberschatz et al., 2020).

Problem with separate files	How a DBMS solves it
Redundancy — the same farmer's details typed in the seed, fertiliser and subsidy registers	Data stored once and shared by all applications
Inconsistency — a mobile number changed in one file but not the others	One update is seen everywhere
Difficult access — a new question needs a new program or hours of manual search	Any question can be answered with a query
Integrity problems — impossible values (pH of 25, negative area) are accepted	Constraints reject invalid data
Concurrent-use problems — two clerks editing the same record at once	Concurrency control keeps updates correct
Weak security — anyone who opens the file sees everything	Each user gets only the rights they need
Poor recovery — data lost when a computer fails	Backup, logging and recovery are built in

3. Three-schema architecture

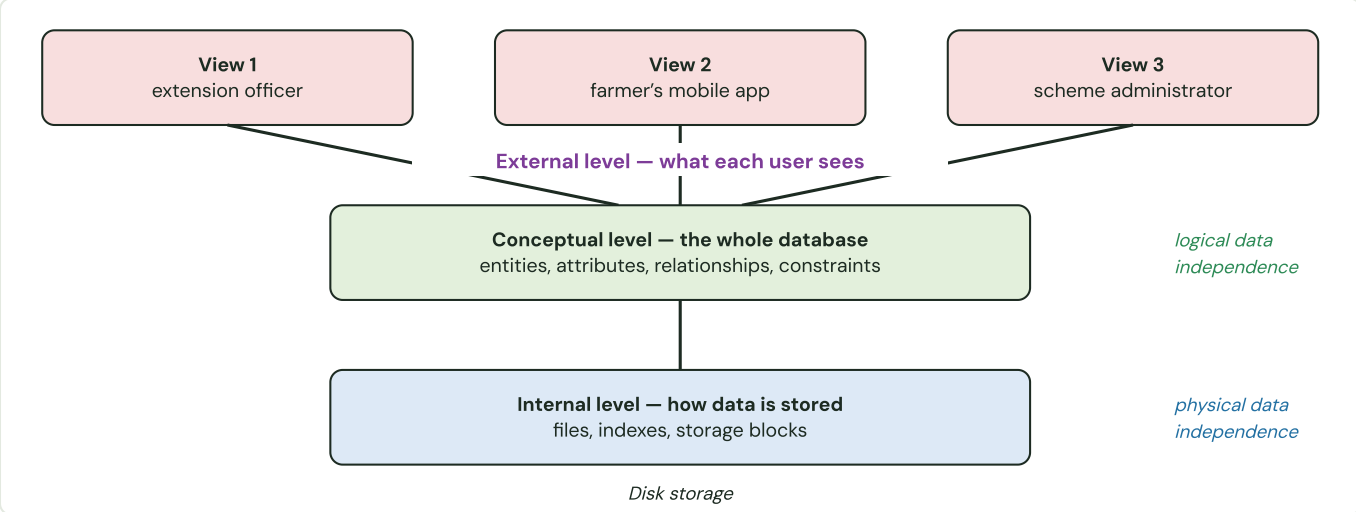


Figure 1. The three levels of a database (ANSI/SPARC architecture). Original diagram.

- **External level** — the different views that different users see (an extension officer sees farmers of her block; a farmer sees only his own record).
- **Conceptual level** — the complete logical structure of the database: entities, attributes, relationships and rules.
- **Internal level** — how the data is physically stored on disk: files, indexes and blocks.
- **Data independence** — a change at one level need not change the levels above it. **Physical** independence: storage can be reorganised without changing the conceptual design. **Logical** independence: the conceptual design can be extended (a new field added) without rewriting users' views.

4. Types of databases (data models)

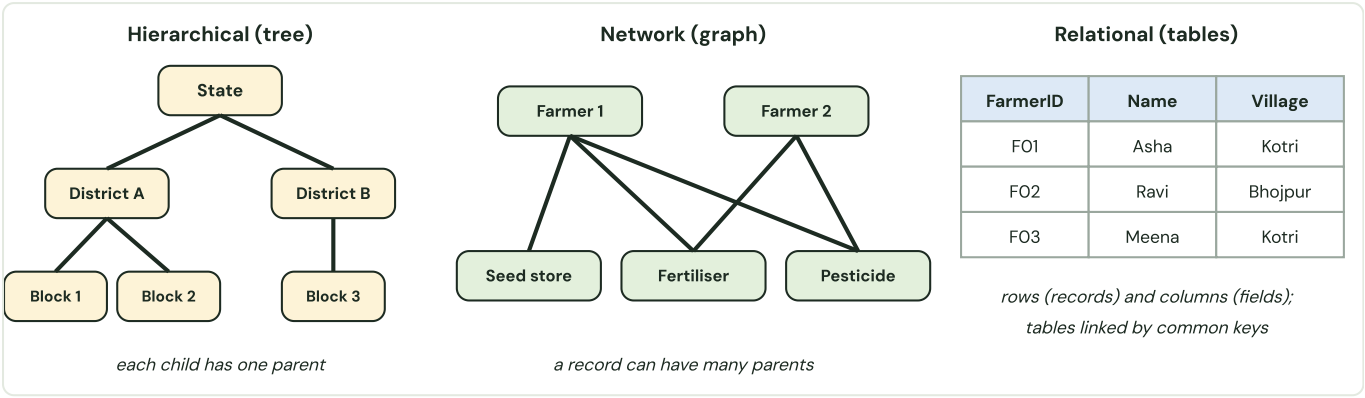


Figure 2. Hierarchical, network and relational ways of organising data. Original diagram.

Model	Structure	Strengths / limits	Examples
Hierarchical	Tree — each child record has exactly one parent	Fast for fixed one-to-many data; poor for many-to-many links	IBM IMS (1960s); folder trees; State → district → block
Network	Graph — a record may have several parents	Handles many-to-many links; complex to design and change	CODASYL databases
Relational	Data in tables (relations) linked by common keys	Simple, flexible, queried with SQL; the most widely used model	MS Access, MySQL, PostgreSQL, Oracle, SQL Server, SQLite
Object-oriented	Data stored as objects with attributes and methods	Suits complex data such as designs and multimedia	Object databases; object features in PostgreSQL
NoSQL	Document, key–value, wide-column or graph stores	Scales to very large, fast-changing or unstructured data (sensor streams, images)	MongoDB (document), Redis (key–value), Cassandra (wide-column), Neo4j (graph)

Databases are also classed by **number of users** (single-user or multi-user) and **location** — **centralised** (one site), **distributed** (parts at several sites linked by a network) or **cloud** (hosted by a provider and reached through the Internet).

5. The relational model

The relational model was proposed by E. F. Codd of IBM in 1970 (Codd, 1970). It represents all data as **relations** — tables of rows and columns — and is the basis of almost every database you will use.

Formal term	Common term	Meaning
Relation	Table	A set of rows about one kind of thing (e.g. FARMER)
Tuple	Row / record	One item — one farmer
Attribute	Column / field	One property — Name, Village
Domain	Allowed values	e.g. pH between 0 and 14
Degree	—	Number of columns
Cardinality	—	Number of rows

Keys

Key	Meaning	Example
Super key	Any set of columns that identifies each row uniquely	{FarmerID, Name}
Candidate key	A minimal super key	FarmerID; also a registry-issued Farmer ID
Primary key	The candidate key chosen to identify rows; never empty or repeated	FarmerID
Foreign key	A column that refers to the primary key of another table, linking the two	FarmerID in the PLOT table
Composite key	A key made of two or more columns	{PlotID, Season} in a cropping table

- **Entity integrity** — a primary key can never be empty (NULL).
- **Referential integrity** — a foreign key must match an existing primary key (a plot cannot belong to a farmer who is not in the FARMER table).

Entity–relationship (ER) diagrams

Before building tables, designers draw an **ER diagram**: **entities** (things, shown as rectangles), their **attributes** (ovals) and the **relationships** between them (diamonds), with **cardinality** — one-to-one (1:1), one-to-many (1:M) or many-to-many (M:N).

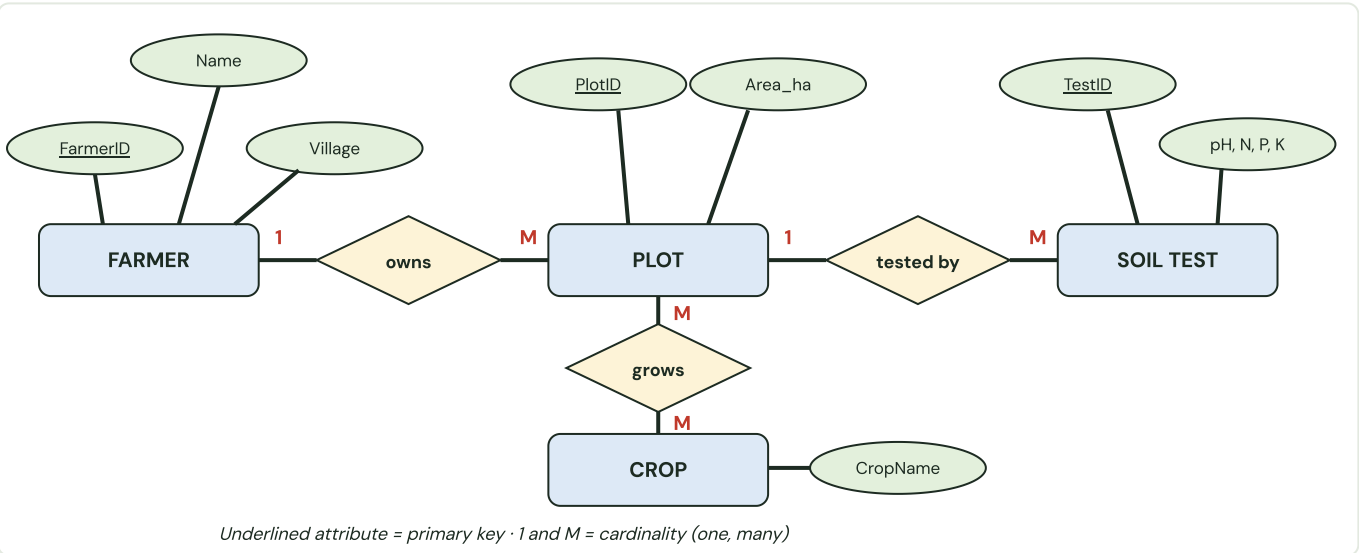


Figure 3. ER diagram of a simple farm database: a farmer owns many plots, each plot has many soil tests, and plots and crops are linked many-to-many. Original diagram.

In tables, a 1:M relationship is made with a foreign key in the "many" table (FarmerID in PLOT). An M:N relationship needs a separate linking table — e.g. CROPPING (PlotID, Season, CropName).

6. Normalisation

Normalisation organises data into well-structured tables to remove repetition and the errors it causes — **update anomalies** (a change must be made in many rows), **insertion anomalies** (a new item cannot be added without unrelated data) and **deletion anomalies** (deleting one fact loses another).

Normal form	Rule
First (1NF)	Every cell holds a single value; no repeating groups
Second (2NF)	1NF, and every non-key column depends on the whole primary key (not part of a composite key)
Third (3NF)	2NF, and no non-key column depends on another non-key column (no transitive dependency)

Worked example — normalising a soil-test register
Unnormalised sheet: TestID | FarmerName | Village | Mobile | PlotArea | TestDate | pH | N
Problems: a farmer with five tests has his name, village and mobile repeated five times; changing his mobile needs five edits; and a farmer cannot be registered until a soil test is done.
Normalised design (3NF):
FARMER (**FarmerID**, Name, Village, Mobile)
PLOT (**PlotID**, FarmerID → FARMER, Area_ha)
SOIL_TEST (**TestID**, PlotID → PLOT, TestDate, pH, N_kg_ha, P_kg_ha, K_kg_ha)
Each fact is now stored once, and the tables are joined by their keys when needed.

7. SQL — the language of relational databases

SQL (Structured Query Language) is the standard language for defining and using relational databases; it is standardised as ISO/IEC 9075 (International Organization for Standardization, 2023). Its commands fall into five groups:

Group	Purpose	Commands
DDL — data definition	Create and change the structure	CREATE, ALTER, DROP
DML — data manipulation	Add, change and remove rows	INSERT, UPDATE, DELETE
DQL — data query	Retrieve data	SELECT
DCL — data control	Give or remove permissions	GRANT, REVOKE
TCL — transaction control	Confirm or undo a group of changes	COMMIT, ROLLBACK

Worked example — SQL on a soil-test database

```
CREATE TABLE SOIL_TEST (TestID INT PRIMARY KEY, PlotID INT, TestDate DATE, pH DECIMAL(3,1), N_kg_ha INT);
INSERT INTO SOIL_TEST VALUES (101, 11, '2026-05-02', 6.2, 240);
-- Plots with low available nitrogen (below 280 kg/ha) and acidic soil:
SELECT PlotID, pH, N_kg_ha FROM SOIL_TEST WHERE N_kg_ha < 280 AND pH < 6.5 ORDER BY N_kg_ha;
-- Average nitrogen in each village (joining three tables):
SELECT F.Village, AVG(S.N_kg_ha) FROM FARMER F JOIN PLOT P ON F.FarmerID = P.FarmerID
JOIN SOIL_TEST S ON P.PlotID = S.PlotID GROUP BY F.Village;
-- Correct a wrongly entered value:
UPDATE SOIL_TEST SET pH = 7.2 WHERE TestID = 101;
```

The 280 kg/ha cut-off follows the common Indian rating of available soil nitrogen (low below 280, medium 280–560, high above 560 kg/ha); it is used here only to illustrate a query.

8. Transactions, security and privacy

- A **transaction** is a group of operations treated as one unit — e.g. "deduct 2 bags of urea from stock and record the sale to farmer F02". Either all of it happens or none of it does.

- Transactions follow the **ACID** properties: **Atomicity** (all or nothing), **Consistency** (rules always hold), **Isolation** (simultaneous transactions do not interfere) and **Durability** (once committed, changes survive failures).
- **Security:** user accounts, passwords and privileges (GRANT, REVOKE); encryption of sensitive data; audit logs.
- **Backup and recovery:** regular backups and transaction logs let the DBMS restore data after a crash.
- **Privacy:** farmer databases hold personal data such as names, mobile numbers, land and bank details. In India, the Digital Personal Data Protection Act, 2023 governs such data, and its Rules were notified in November 2025 for phased implementation — collect only the data needed, obtain consent and protect it.

9. Popular DBMS software

DBMS	Type	Typical use
MS Access	Desktop relational, with forms and reports	Small office databases — a cooperative's member records
MySQL / MariaDB	Open-source relational server	Websites and portals
PostgreSQL (with PostGIS)	Open-source relational server; PostGIS adds maps and spatial queries	GIS databases of fields and villages (Lecture 15)
Oracle Database, Microsoft SQL Server	Commercial relational servers	Large government and business systems
SQLite	Small embedded relational database inside an app	Offline storage in mobile farm apps
MongoDB	NoSQL document database	Flexible records such as sensor and app data

MS Access in brief

Object	Purpose
Tables	Store the data (Datasheet view to enter data; Design view to set fields, data types and the primary key)
Queries	Ask questions of the data and combine tables
Forms	Friendly screens for entering and viewing records
Reports	Formatted printouts, e.g. a soil-test summary per village
Macros and modules	Automate tasks

Common field data types are Short Text, Long Text, Number, Date/Time, Currency, Yes/No and AutoNumber (automatic serial numbers, useful as primary keys).

10. Databases in agriculture

Database or platform	What it holds and why it matters
Soil Health Card portal (soilhealth.dac.gov.in)	Soil-test results and fertiliser recommendations for farmers' fields. The scheme was launched on 19 February 2015 at Suratgarh, Rajasthan, and the portal generates the cards from laboratory data (Press Information Bureau [PIB], 2025)
AgriStack (Digital Agriculture Mission)	Approved by the Union Cabinet on 2 September 2024 with an outlay of ₹2,817 crore. Its foundational registries are the Farmers' Registry, Geo-referenced Village Maps and the Crop Sown Registry, managed by states and union territories (PIB, 2024) — Lecture 31
Livestock databases (Bharat Pashudhan)	Animal identification, breeding, vaccination and treatment records linked to Pashu Aadhaar ear tags (Lecture 2)
Market-price databases	Daily arrivals and prices in regulated markets, used by price apps and policy makers (Lecture 14)
Weather and climate databases	Station and gridded rainfall and temperature data used for advisories and crop models (Lectures 11, 13)
Research databases	Field-trial results, germplasm collections and soil and water survey data in ICAR institutes and universities
Farm and agri-business records	Input stock, machinery logs, dairy and cooperative member records, often in MS Access, spreadsheets or app databases

Advantages and limitations of a DBMS

Advantages	Limitations
Less redundancy and inconsistency	Cost of software, hardware and trained staff
Data shared by many users and applications	Complex to design well
Integrity constraints and security	A central failure affects every user unless well backed up
Powerful queries with SQL	Needs connectivity for shared online databases
Backup, recovery and concurrency control	Privacy risks if personal data is poorly protected

11. Key takeaways

- A DBMS stores data once, shares it safely and answers questions through queries — solving the redundancy, inconsistency and security problems of separate files.
- The three-schema architecture separates users' views, the logical design and physical storage, giving data independence.
- Data models include hierarchical, network, relational, object-oriented and NoSQL; the relational model (Codd, 1970) dominates.
- Tables are linked by primary and foreign keys; ER diagrams show entities, relationships and cardinality.
- Normalisation (1NF → 2NF → 3NF) removes repetition and anomalies.
- SQL defines (DDL), changes (DML), queries (SELECT), secures (DCL) and controls transactions (TCL); transactions follow ACID.

- Indian agriculture increasingly runs on databases — Soil Health Card, AgriStack registries, livestock and market databases — so data quality and privacy matter.

12. Examination questions

Short-answer questions (2 marks)

Q1. Define DBMS. Give two examples.

Software to create, store, retrieve, update and protect databases — e.g. MySQL, MS Access.

Q2. What is data redundancy?

Storing the same data in more than one place, which wastes space and leads to inconsistency.

Q3. Name the three levels of the ANSI/SPARC architecture.

External, conceptual and internal.

Q4. Distinguish between physical and logical data independence.

Physical: storage can change without changing the logical design. Logical: the logical design can change without changing users' views.

Q5. Define primary key and foreign key.

A primary key uniquely identifies each row; a foreign key refers to the primary key of another table.

Q6. Who proposed the relational model, and when?

E. F. Codd of IBM, in 1970.

Q7. What is normalisation?

Organising data into well-structured tables to remove redundancy and update, insertion and deletion anomalies.

Q8. Expand DDL and DML and give one command of each.

Data definition language (CREATE) and data manipulation language (INSERT).

Q9. What does ACID stand for?

Atomicity, consistency, isolation and durability.

Q10. Name the three foundational registries of AgriStack.

Farmers' Registry, Geo-referenced Village Maps and Crop Sown Registry.

Long-answer questions (5–10 marks) — answer outlines

Q1. Compare file-based systems and DBMS.

Problems of files (redundancy, inconsistency, access, integrity, concurrency, security, recovery) and how a DBMS solves each (table in Section 2).

Q2. Explain the three-schema architecture with a diagram.

Figure 1; the three levels; physical and logical data independence with examples.

Q3. Describe the types of data models.

Hierarchical, network, relational, object-oriented and NoSQL — structure, strengths, examples (Figure 2 and table).

Q4. Explain relational terms, keys and integrity constraints.

Relation, tuple, attribute, domain, degree, cardinality; super, candidate, primary, foreign and composite keys; entity and referential integrity.

Q5. Draw an ER diagram for a farm database and convert it into tables.

Figure 3; FARMER, PLOT, SOIL_TEST, CROP; 1:M via foreign keys; M:N via a linking table; normalised design.

Q6. Discuss the applications of databases in agriculture.

Soil Health Card, AgriStack, livestock, market, weather and research databases; farm and agri-business records; privacy under the DPDP Act, 2023.

13. Glossary

Term	Meaning
ACID	Atomicity, consistency, isolation, durability — properties of reliable transactions
Attribute	A column (field) of a table
Cardinality	Number of rows in a table; or one/many in a relationship
Database	An organised collection of related, shared data
DBA	Database administrator
DBMS	Database management system
Domain	The set of allowed values for an attribute
ER diagram	Entity–relationship diagram showing entities, attributes and relationships
Foreign key	A column referring to the primary key of another table
NoSQL	Non-relational databases: document, key–value, wide-column, graph
Normalisation	Structuring tables to remove redundancy and anomalies
Primary key	The column(s) uniquely identifying each row
Query	A request for data from a database
Relation	A table in the relational model
Schema	The structure (design) of a database
SQL	Structured Query Language for relational databases
Transaction	A group of operations executed as one unit
Tuple	A row (record) of a table

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All diagrams are original. Explanations are written in original language; standard content on database architecture, models, keys, normalisation and SQL follows Elmasri and Navathe (2016) and Silberschatz, Korth and Sudarshan (2020). Farm tables and SQL data are illustrative.

Practice MCQs

Q1. DBMS stands for

(a) Data Backup Management System
(b) Database Management System
(c) Digital Base Memory System
(d) Data Bus Management System

Q2. Storing the same data in several files is called

(a) integrity
(b) redundancy
(c) normalisation
(d) independence

Q3. The ANSI/SPARC level that describes physical storage is the

(a) external level
(b) conceptual level
(c) internal level
(d) view level

Q4. In a hierarchical database each child record has

(a) no parent
(b) exactly one parent
(c) many parents
(d) two parents

Q5. The relational model was proposed by

(a) Charles Babbage
(b) E. F. Codd
(c) John von Neumann
(d) Tim Berners-Lee

Q6. A row of a relational table is also called a

(a) tuple
(b) domain
(c) attribute
(d) schema

Q7. A column that refers to the primary key of another table is a

(a) candidate key
(b) super key
(c) foreign key
(d) composite key

Q8. A many-to-many relationship is implemented using

(a) one table
(b) a linking table
(c) a primary key only
(d) a view

Q9. Which SQL command retrieves data?

(a) INSERT
(b) UPDATE
(c) SELECT
(d) CREATE

Q10. GRANT and REVOKE belong to

(a) DDL
(b) DML
(c) DCL
(d) TCL

Q11. In ACID, "A" stands for

(a) Accuracy
(b) Atomicity
(c) Availability
(d) Authentication

Q12. Which is NOT one of AgriStack's foundational registries?

(a) Farmers' Registry
(b) Crop Sown Registry
(c) Geo-referenced Village Maps
(d) Mandi Price Registry

Answers

Q1 — (b) Database Management System

Q2 — (b) redundancy
Q3 — (c) internal level
Q4 — (b) exactly one parent
Q5 — (b) E. F. Codd
Q6 — (a) tuple
Q7 — (c) foreign key
Q8 — (b) a linking table
Q9 — (c) SELECT
Q10 — (c) DCL
Q11 — (b) Atomicity
Q12 — (d) Mandi Price Registry